

Mallikarjun Reddy

AI/ML ENGINEER

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👤 PROFESSIONAL SUMMARY

AI/ML Engineer with expertise in **Natural Language Processing, Computer Vision, and Generative AI**. Proven track record in delivering **enterprise-grade conversational AI platforms, RAG systems**, and scalable **distributed data pipelines (Kafka, PySpark)**. Skilled in finetuning open-source LLMs (**Llama, Qwen, DeepSeek**) and deploying optimized GPU workloads. Passionate about bridging AI research and real-world applications.

🏢 EXPERIENCE

AI/ML Engineer

Aug 2024 – Present

Isourse Technologies

- Built a **conversational BI platform** enabling SQL database interaction via natural language queries, replacing manual dashboard creation.
- Developed a **RAG-based support chatbot** with **persistent agentic memory** to improve contextual conversations and user retention.
- Finetuned & deployed open-source LLMs (**Llama, Qwen, DeepSeek**) for Text-to-SQL and chatbot solutions using **SFT, PEFT, and LoRA**.
- Orchestrated **real-time data warehousing pipeline** (APIs → Kafka → PySpark) to process high-volume distributed data streams.
- Designed an **Intelligent ATS system** leveraging **vector embeddings** and semantic similarity for automated candidate logical scoring.
- Migrated legacy ML services to **Dockerized FastAPI microservices**, implementing monitoring via **Grafana & Prometheus**.
- Implemented **YOLO + OCR** for automated document text extraction and **FaceNet** for secure user biometric authentication.

ML Researcher

Oct 2023 – Aug 2024

Independent / Freelance

- Developed autonomous **Chess-Bot** using custom ML evaluator and optimized **MiniMax algorithm**.
- Engineered hybrid **Air Quality Prediction model** ensemble (LSTM, GRU, Prophet) achieving **lower RMSE** than single-model baselines.
- Finetuned **Longformer Encoder-Decoder models** (16k context) for long-document summarization using **QLoRA** efficiency methods.

Co-Founder & Developer

Jul 2022 – Mar 2023

Coupondel (IIT Roorkee Pre-incubated)

- Architected and launched an **e-commerce coupon discovery & cashback platform** with integrated affiliate marketing networks.
- Led a team of **5 developers** to build the MVP and successfully deployed the **production web application**.

<> **SELECTED PROJECTS**

3D Avatar Motion Capture

MediaPipe, Three.js, WebGL

Real-time browser-based 3D avatar system mirroring full body, hand, and face movements via webcam.

Enterprise AI Chatbot

Gemini 2.0, Llama 3.1, FastRTC

Next-gen corporate assistant featuring hybrid search (keyword + semantic) and real-time voice interaction.

Identity Verification API

YOLO, OCR, FastAPI, ONNX

High-throughput CPU-optimized Computer Vision API for extracting data from Indian Identity Documents.

BI Analytics Backend

Text-to-SQL, LLMs, Docker

Backend service converting natural language questions into executable SQL queries for business intelligence.

VisualML Video Generator

Manim, Remotion, Gemini API

Automated pipeline generating short educational videos from text prompts using programmatic animation tools.

Face Recognition System

InsightFace, ONNX Runtime

Asynchronous, high-performance face recognition microservice optimized for low-latency inference.

Logistics Data Pipeline

Spark, Kafka, PostgreSQL

Robust ELT pipeline for ingesting and transforming logistics data streams for analytics.

Intelligent ATS API

FastAPI, Gemini 2.0

Resume parsing and scoring API comparing candidate profiles against job descriptions.

 TECHNICAL SKILLS

Languages & Frameworks: Python, C++, SQL, FastAPI, PyTorch, TensorFlow, LangChain, Node.js, React

AI/ML: LLMs (Llama/Qwen/DeepSeek), RAG, Agentic Workflows, Text-to-SQL, OCR, CNNs, Transformers, LoRA/PEFT, MediaPipe, YOLO, InsightFace

Data & Cloud: Kafka, PySpark, PostgreSQL, Milvus, ClickHouse, AWS, GCP, Azure, Docker, Kubernetes, Grafana, Prometheus

Tools: Git, VS Code, Linux, CI/CD, ONNX Runtime, Streamlit, FFmpeg

 EDUCATION

B.Tech, Mechanical Engineering

2019 – 2023

Indian Institute of Technology Roorkee

CGPA: 7.4/10 • JEE Advanced Rank: 3015

 CERTIFICATIONS

Machine Learning Specialization (Coursera/Stanford)

Web Developer Bootcamp (Udemy)

Deep Learning Specialization (DeepLearning.AI)

Competitive Programming (NPTEL)